

ICAD: Rethinking the Role of Inference and Cues for the Anomaly Detection of Time Series in IIoT

Zhichao Wu, Li Zhu, Zitao Yin, Shunqi Zhang, Xirong Xu, Xiaopeng Wei, and Xin Yang

Abstract—Time series anomaly detection in real-world Industrial Internet of Things (IIoT) systems is pivotal for identifying unsafe conditions and implementing timely preventive measures. While diffusion models are popular for capturing complex patterns, they often struggle to balance diversity and fidelity across scenarios due to limited exploration of context-window logical inference relationships and trend-pattern cues. To address these challenges, we propose ICAD, a novel method that rethinks the role of inference and cues in IIoT time series anomaly detection. ICAD defines trend patterns and uncertainties in textual form and utilizes a fine-tuned large language model to encode these descriptions as conditions for the diffusion model, thereby enhancing its generalization across diverse data distributions. Additionally, a “reasoning network for contextual window” mechanism is designed to capture temporal dependencies between adjacent windows, complemented by multi-scale and spatial feature adaptive fusion modules to further enhance the predictive performance. Empirical evaluations across four benchmark datasets and a large-scale ethylene oxide production process demonstrate that ICAD consistently outperforms state-of-the-art baselines, confirming its effectiveness and practicality in overcoming current anomaly detection model limitations.

Index Terms—Inference and Cues, Anomaly Detection, Conditional Diffusion model, IIoT.

I. INTRODUCTION

ANOMALY detection is essential in the Industrial Internet of Things (IIoT) due to the increasing reliance on interconnected sensors, instruments, and devices. From one perspective, many of these devices are designed with limited security features, exposing them to potential exploitation by malicious actors. Efficient anomaly detection is crucial for ensuring the security and integrity of IIoT systems, safeguarding industrial operations. From another perspective, using real-time industrial data, anomaly detection can identify abnormal situations not only as they occur but also proactively before they manifest. With effective anomaly detection, the risks of economic losses from system failures, safety incidents, or downtime can be significantly mitigated [1], [2].

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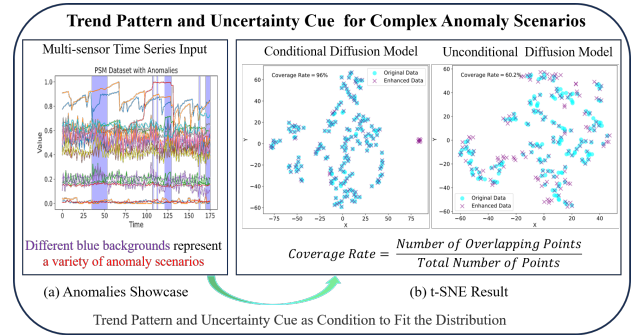


Fig. 1. Trend pattern and uncertainty cue for complex anomaly scenarios. (a) shows the original time series along with their corresponding anomaly indicators. (b) compares the dimensionality-reduced distributions of the time series generated under clue-conditioned and clue-free scenarios.

Due to the inherent scarcity of labeled anomalies and the diverse nature of anomaly types, anomaly detection is typically approached as an unsupervised learning task [3], [4]. These techniques can be broadly classified into two main categories: prediction-based [5]–[8] and reconstruction-based [4], [9], [10]. Unlike prediction-based methods, reconstruction-based methods are widely used for their ability to avoid the cumulative effect of prediction errors on anomaly detection tasks [11].

Diffusion models, an emerging and promising reconstruction technique, have shown superior performance compared to traditional generative models such as Autoencoders (AEs), Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs) in time series anomaly detection [12]–[17]. This stems from the complexity of IIoT data, characterized by multi-modal distributions, rare events, and dynamic variations. The mathematical framework of diffusion models enables them to approximate multi-distribution scenarios through iterative denoising, making them particularly effective for anomaly detection in such environments [18], [19]. However, current diffusion models face challenges in balancing time series fidelity and diversity in complex scenarios, impeding their practical deployment in real-world production processes [20], [21]. Specifically:

- The failure to account for inherent trend patterns and uncertainty descriptions results in excessive variability in the generated data, limiting the model’s ability to accurately fit the true distribution of anomaly scenarios. Specifically, the complexity of anomaly distributions in IIoT stems from the system’s high interconnectivity. Minor perturbations, such as external disturbances or equip-

ment degradation, can cause abrupt shifts in operational conditions, leading to uncertain and diverse anomaly patterns (highlighted in purple on the left side of Fig. 1). By incorporating these inherent factors, the conditional diffusion model outperforms the unconditional model by effectively constraining this variability. This enables the conditional model to better capture the underlying trend patterns and uncertainty, leading to a higher coverage rate (as shown on the right side of Fig. 1), improved generalization, and enhanced prediction accuracy.

- During the process of adding noise to time series window data, diffusion models often lose the inherent logical relationships between contextual windows. Naturally, this disruption can ultimately compromise the model's performance.

In this paper, we rethink the role of inference and cues for IIoT time series anomaly detection, and propose a novel method, termed ICAD, to address the aforementioned issues. Firstly, a textual description of trend patterns and uncertainties is defined as input to the diffusion model. Subsequently, a logic inference network based on contextual windows is proposed to focus on the important information and ensure the fidelity of the generated samples. Finally, we propose a multi-scale and spatial adaptive fusion module to train the denoising network.

Overall, the main contributions of our work are summarized as follows:

- *Enhanced Generalization*: To enhance the generalization ability of the diffusion model across diverse abnormal data distributions, we fine-tune a pre-trained large language model, BERT, to encode defined textual descriptions of trend patterns and uncertainties. This is verified by the generalization test results presented in Subsection 5.8.
- *Contextual Inference Network*: An inference network is defined for contextual windows to address the diffusion model's neglect of relationships between these windows, ensuring its ability to detect both progressive and concealed anomalies.
- *Adaptive Fusion Module*: We design an adaptive fusion module for multi-scale and spatial features, serving as the denoising network in the diffusion model, which enhances the model's capability to capture complex anomaly characteristics in challenging scenarios.
- *Comprehensive Evaluation*: Experiments are conducted on five real-world multivariate time series datasets, including four existing datasets and a newly collected dataset from a large-scale fine chemical enterprise. These datasets span various IIoT domains. A comparative analysis with 23 baselines shows that ICAD significantly outperforms state-of-the-art methods. We publicly publish our code and EO dataset of experiments on GitHub¹ for better reproducibility of the results of this paper.

II. RELATED WORK

A. Multivariate Time Series Anomaly Detection

With the rapid growth of interconnected devices and sensors in cyber-physical systems, multivariate time series (MTS)

anomaly detection has received significant attention from the industrial communities, leading to the development of various algorithms [22]–[24].

Classical methods, such as Local Outlier Factor and variations [25], [26], Isolation Forest [27], Empirical-Cumulative-distribution-based Outlier Detection [28], etc., do not explicitly model the contextual information or deep feature representations within MTS data. Consequently, they struggle to adapt to complex industrial sensor detection tasks.

In recent years, deep learning algorithms have made breakthroughs in MTS anomaly detection due to their powerful feature representation capabilities, typically falling into forecasting-based or reconstruction-based categories [29].

Forecasting-based methods determine the anomaly scores of the samples by assessing the disparity between the model predictions and input data. Graph Deviation Network (GDN) uses graph neural networks with an attention mechanism to capture sensor topologies for anomaly detection [5]. Lunar employs information from the nearest neighbors of each node in a trainable manner to detect anomalies [6].

Conversely, reconstruction-based methods exploit the difficulty of learning abnormal samples, resulting in relatively large errors when reconstructing the original samples. USAD trains autoencoders to reconstruct the original samples [4]. ANOMALY TRANSFORMER computes the association discrepancies by a new Anomaly Attention mechanism [30]. COUTA realizes contamination-tolerant, anomaly-informed learning of data normality via uncertainty modeling-based calibration and native anomaly-based calibration [31].

However, current deep learning algorithms often struggle to efficiently generate diverse time series data, making it challenging to capture complex patterns and potential data distributions. This limitation hinders their ability to adapt to complex scenarios, especially when anomalies are intricate or abnormal samples are scarce.

B. Diffusion Model

The application of diffusion models, a powerful generative model, in the field of time series is relatively new, and researchers are beginning to explore its potential. DiffAD enhances the imputation performance of missing values by a new denoising diffusion-based method with conditional weight-incremental diffusion [13]. DDMT integrates a novel adaptive dynamic neighbor mask mechanism with Transformer and DDPM models for anomaly detection [14]. ImDiffusion combines time series imputation and diffusion models to achieve accurate and robust anomaly detection [15]. Diffstg integrates the spatio-temporal learning capabilities of spatio-temporal graph neural networks with the uncertainty measurements of diffusion models [32].

To the best of our knowledge, our proposed ICAD represents the pioneering application of the textual descriptions of trend patterns and uncertainties as diffusion model conditions and the contextual window logic in anomaly detection with MTS.

¹Code and Dataset: <https://github.com/Anomaly-Detection3/ICAD>

III. PRELIMINARIES

A. Multivariate Time Series Anomaly Detection

Given multivariate time series $X = \{X_i^j\}$, $i \in [1, 2, \dots, N]$, $j \in [1, 2, \dots, M]$, where M and N are the maximum length of timestamps and the number of sensors in the input, respectively. The goal of MTS anomaly detection is to identify whether an observation deviates significantly from the overall pattern of X , indicating an anomaly. To capture the local contextual information, a sliding window of length w over the MTS is adopted to obtain the result X_W , defined as:

$$X_W = \{W^1, W^2, \dots, W^k\}, \quad (1)$$

where k , calculated as $\lfloor \frac{N}{w} \rfloor$, is the number of windows.

B. Denoising Diffusion Probabilistic Model

The diffusion model consists of two processes: the forward process and the backward process. During the forward process, noise is incrementally added to the input X_0 , ultimately transforming it into Gaussian noise X_T after T steps. Conversely, the backward process generates samples by progressively removing the noise.

The forward process $q(X_t|X_{t-1})$ aims at obtaining X_t by adding Gaussian noise to the time series X_{t-1} at each step t , $t \in [1, 2, \dots, T]$. Therefore, mathematically, the expression for the forward process is as follows:

$$q(X_t|X_{t-1}) = N(X_t; \sqrt{1 - \beta_t}X_{t-1}, \beta_t\mathbf{I}), \quad (2)$$

where β_t is a fixed variance linearly increasing with t , representing the noise level.

During the backward process, the noise is gradually removed to generate the samples, expressed as:

$$p_\theta(X_{t-1}|X_t) = N(X_{t-1}; \mu_\theta(X_t, t), \beta_t\mathbf{I}). \quad (3)$$

Instead of learning to predict $\mu_\theta(X_t, t)$, a more efficient approach trains a network ϵ_θ to predict the noise ϵ . The training loss is defined as:

$$\min_\theta \mathcal{L}'(\theta) = \min_\theta E_{x_0 \sim q(x_0), \epsilon \sim N(0, \mathbf{I}), t} \|\epsilon - \epsilon_\theta(X_T, t)\|_2^2. \quad (4)$$

IV. ICAD METHOD

The overall architecture of the proposed ICAD method, depicted in Fig. 2, adheres to the foundational structure of DDPM [33]. ICAD comprises three modules: Textual Embedding for Trend Cue Description, Reasoning Network for Contextual Window Logic, and Model Training and Anomaly Inference.

- *Textual Embedding for Trend Cue Description*: By defining trend patterns and converting time series into textual cues, we utilize a pre-trained BERT model [34] to encode cues into text embeddings. These encoded vectors serve as conditions during the backward process of the diffusion model. This strategy ensures sample fidelity while promoting diversity in the samples generated by DDPM.
- *Reasoning Network for Contextual Window Logic*: We ingeniously transform traditional logical reasoning rules

into a format suitable for neural network training. Specifically, we define the logical relationships between contextual windows and express these relationships through contrastive learning using positive and negative samples. This approach enables the diffusion model to better understand and capture the logical connections between contextual windows, thereby significantly enhancing the model's reasoning capabilities.

- *Model Training and Anomaly Inference*: To enhance the stability of anomaly detection results, we jointly optimize the Reasoning Network and conditional diffusion model. For training the diffusion model, we devise two key techniques: 1) employing a multi-scale feature and spatial feature adaptive fusion module to train the denoising network of the diffusion model, and 2) utilizing a Kalman filter [35] to perceive the state space values of the time series, which are then used as inputs for the forward process of the diffusion model.

To elucidate the proposed ICAD method, we present the pseudo-code of its workflow in Algorithm 1 and 2. The algorithm comprises two primary phases: training and reasoning. During the training phase, noise is introduced to the original data, and the model is trained to predict this noise. In the reasoning phase, the trained denoising model is used to sample and reconstruct the noise, thereby clarifying the underlying data structure.

A. Textual Embedding for Trend Cue Description

Effectively capturing trend cues is crucial for enhancing the generalization capacity of models in IIoT applications. Conventional methods, such as trend decomposition, excel at extracting numerical trends but often treat them as isolated features, overlooking the role of qualitative insights in anomaly detection. Humans assess anomalies by combining quantitative measurements with qualitative reasoning about patterns, often summarizing these observations as descriptive narratives. Inspired by this, we propose a mechanism that integrates both qualitative and quantitative descriptions of trend patterns as priors, structured to guide anomaly detection. The proposed ICAD method adapts to diverse and dynamic operating conditions by automatically generating trend-aware representations, thereby facilitating the early recognition of operational shifts.

Given the window input $W^d \in R^{N \times w}$, $d \in [1, 2, \dots, k]$, the first-order difference is computed as:

$$\Delta X_{i,j} = W_{i,j+1}^d - W_{i,j}^d, \quad (5)$$

where $i = 1, 2, \dots, N$ and $j = 1, 2, \dots, w - 1$.

After obtaining ΔX , we define the trend pattern and information entropy to characterize trend cues.

Definition 1: Trend Pattern

For the first-order difference ΔX of a window, we firstly partition ΔX into two distinct intervals, $\Delta X^{(1)}$ and $\Delta X^{(2)}$. For each feature within these intervals, we calculate the mean $\mu_i^{(1)}$, $\mu_i^{(2)}$ and variance $\sigma_i^{(1)}$, $\sigma_i^{(2)}$ as:

$$\mu_i^{(1)} = \frac{1}{q} \sum_{p=1}^q \Delta X_{i,p}^{(1)}, \quad \mu_i^{(2)} = \frac{1}{q} \sum_{p=q}^{2q-1} \Delta X_{i,p}^{(2)}, \quad (6)$$

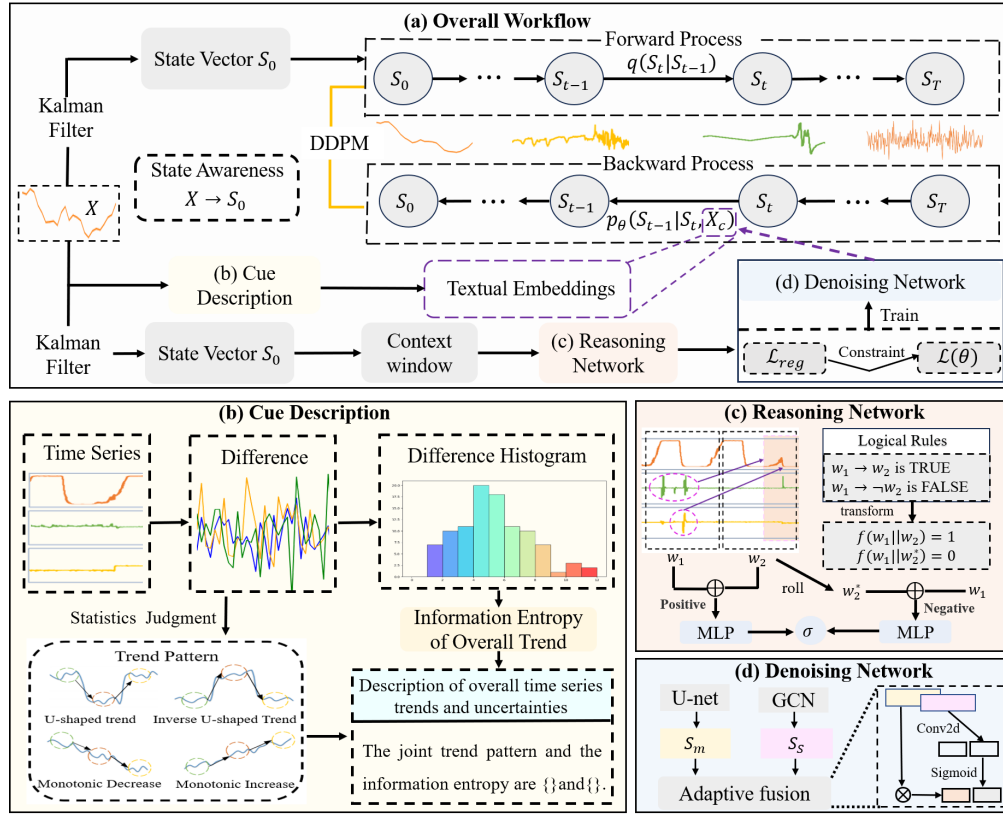


Fig. 2. The framework of the proposed ICAD method. The upper module is the flow chart of ICAD, the lower module is the detailed explanations of Cue Description, Reasoning Network and Denoising Network, respectively. The Cue Description module firstly processes the input X using differential calculations and predefined statistical methods to automatically determine the trend pattern. Subsequently, it calculates the joint entropy of histograms to generate textual cues describing the trend. The Reasoning Network module constructs a contrastive learning network to simulate logical relationships within a contextual window, forming the inference loss function. The Denoising Network module adaptively fuses multi-scale features with the spatial features obtained by Unet and GCN with the attention mechanism.

$$\sigma_i^{(1)} = \frac{1}{q} \sum_{p=1}^q (\Delta X_{i,p}^{(1)} - \mu_i^{(1)})^2, \quad \sigma_i^{(2)} = \frac{1}{q} \sum_{p=q}^{2q-1} (\Delta X_{i,p}^{(2)} - \mu_i^{(2)})^2, \quad (7)$$

where $q = \lfloor (w-1)/2 \rfloor$ and $i = 1, 2, \dots, N$.

Subsequently, we compute the ratio of the mean to the variance, denoted as $M_i^{(1)}$ and $M_i^{(2)}$, for each feature, as following:

$$M_i^{(1)} = \frac{\mu_i^{(1)}}{\sigma_i^{(1)}}, \quad M_i^{(2)} = \frac{\mu_i^{(2)}}{\sigma_i^{(2)}}. \quad (8)$$

This ratio quantifies the degree of trend deviation, providing a measure of the stability and variability of the time series data across the two intervals. To reflect the overall trend of the feature set, the mean values $M^{(1)}$ and $M^{(2)}$ are computed by averaging these ratios across all feature dimensions, expressed as:

$$M^{(1)} = \frac{1}{N} \sum_{i=1}^N M_i^{(1)}, \quad M^{(2)} = \frac{1}{N} \sum_{i=1}^N M_i^{(2)}. \quad (9)$$

Finally, the trend pattern is identified using the judgment function f , which determines the pattern type based on the values of $M^{(1)}$ and $M^{(2)}$:

$$f = \begin{cases} \text{U-shaped Trend} & \text{if } M^{(1)} < 0 \text{ and } M^{(2)} > 0 \\ \text{Inverse U-shaped Trend} & \text{if } M^{(1)} > 0 \text{ and } M^{(2)} < 0 \\ \text{Monotonic Decrease} & \text{if } M^{(1)} < 0 \text{ and } M^{(2)} < 0 \\ \text{Monotonic Increase} & \text{if } M^{(1)} > 0 \text{ and } M^{(2)} > 0. \end{cases} \quad (10)$$

Definition 2: Information Entropy

Given the ΔX of a window, a “flatten” operation, compressing the distribution of multidimensional data into a single dimension, is adopted to make it easier to observe and understand the distribution characteristics of overall data distribution, as follows:

$$\phi = \text{flatten}(\Delta X), \quad (11)$$

where $\Delta X \in R^{N \times (w-1)}$ and $\phi \in R^{(Nw-N)}$.

Then, to further analyze the distribution of ϕ , a histogram is constructed as:

$$h_i = \sum_{j=1}^{Nw-N} \delta(\phi_j \in [b_i, b_{i+1})), \quad (12)$$

where h_i and $Nw-N$ are the frequency count for the i -th bin and the total number of samples, respectively. ϕ_j and $[b_i, b_{i+1})$ represent the j -th data point and the boundaries of the i -th bin, with b_i and b_{i+1} marking its start and end points, respectively. The indicator function $\delta(\phi_j \in [b_i, b_{i+1}))$ equals 1 if ϕ_j lies within the i -th bin, and 0 otherwise.

Finally, the trend uncertainty H of the time series is calculated by the histogram information entropy as follows:

$$H = - \sum_{i=1}^l p_i \log(p_i), \quad (13)$$

where l is the number of bins in the histogram, p_i is the probability for the i -th bin, calculated as the ratio of the sample count in that bin to the total number of samples, i.e., $p_i = h_i / (Nw - N)$.

Based on the aforementioned definitions of trend and uncertainty, we design an input template to enhance the comprehension of the large language model. *The template is as follows: "The trend pattern of the time series is {}, the overall uncertainty is {}, please provide a detailed description of the trend information for this window."* Subsequently, a pre-trained BERT model [34] is utilized for text encoding. Finally, an MLP is fine-tuned to generate the conditions necessary for diffusing the model input:

$$X_c = \text{MLP}(\text{Bert}(\text{Text})), \quad (14)$$

where $X_c \in R^{k \times h}$. MLP , Text and h are multi-layer perceptron, the template and embedding dimension, respectively. The visualization of difference histograms and trend patterns is provided by Fig. 5.

B. Reasoning Network for Contextual Window Logic

Anomalies in time series data often manifest as dynamic patterns across adjacent windows, which are difficult to capture through isolated window analysis. Identifying these context-driven dependencies is essential for uncovering hidden patterns and modeling temporal dynamics accurately. To better capture these relationships, we replace raw time series data with state representations that abstract essential trends and filter out irrelevant fluctuations, enabling the identification of logical dependencies between contextual windows.

The proposed reasoning framework models these dependencies by focusing on transitions between state representations. If the state of window A can be inferred from the state of window B , it suggests a logical connection between these windows. Conversely, if such an inference is not possible, no logical relationship exists. The corresponding symbolic logic expression is as follows:

$$S_1 \rightarrow S_2 \text{ is TRUE}, \quad (15)$$

$$S_1 \rightarrow \neg S_2 \text{ is FALSE}, \quad (16)$$

where S_1 and S_2 represent the states of windows A and B , obtained by the Kalman filter [35].

The neural reasoning network is then used to model the symbolic logic and realize the transformation of the above symbolic logic, obtained as:

$$E_+ = \sigma(f(S_1 \parallel S_2)), \quad (17)$$

where σ represents the sigmoid function, and f is composed of two linear layers and a RELU activation function.

Since the reasoning of Eq. (17) is logically reasonable, we consider it to be a positive sample. In order to increase the robustness of the model, we introduce negative samples

$$E_- = \sigma(f(S_1 \parallel \psi(S_2))), \quad (18)$$

where $\psi(\bullet)$ is a random cyclic shift.

Formally, the optimization objective function of positive and negative samples is as below:

$$\mathcal{L}_{reg} = \log(E_+) - \log(1 - E_-). \quad (19)$$

C. Model Training and Anomaly Inference

1) Model Training: Traditional diffusion models lack the capability to incorporate conditional information during the backward process. To address this issue, we use trend text encoding X_c as a condition for the diffusion model. In particular, we use the state [35] of the time series as input to the diffusion model, enabling the model to focus on the underlying patterns, mitigate high-frequency noise, and enhance anomaly detection performance.

The goal of the conditional reverse diffusion process is to infer S_{t-1} by leveraging the information S_t and X_c . The condition diffusion models employ the same parameterization as DDPM, with the only difference being the learning of backward process parameters ϵ_θ , which incorporates additional inputs from the trend cue condition X_c . The process is described as follows:

$$p_\theta(S_{t-1}|S_t, X_c) = N(S_{t-1}; \mu_\theta(S_t, X_c, t), \beta_t I), \quad (20)$$

where $\mu_\theta(S_t, X_c, t)$ is the estimated mean of the conditional backward process, and β_t is a fixed constant. The backward process starts with a noise state vector S_T sampled from a Gaussian noise, and generates samples with similar properties to the original state S_0 by $p_\theta(S_{t-1}|S_t, X_c)$.

To parameterize $\mu_\theta(S_t, X_c, t)$, the multi-feature fusion network (denoising network) is employed to approximate the target $\epsilon_\theta(S_T|X_c, t)$ to predict the noise ϵ by minimizing the following loss function:

$$\min_{\theta} \mathcal{L}(\theta) = \min_{\theta} E_{S_0 \sim q(S_0), \epsilon \sim N(0,1), t} \|\epsilon - \epsilon_\theta(S_T|X_c, t)\|_2^2. \quad (21)$$

To enhance the diffusion model's ability to learn the relationship between contextual windows, we jointly train the denoising network loss and the logical network loss, resulting in the following final optimization function:

$$L_{total} = \mathcal{L}(\theta) + \mathcal{L}_{reg}. \quad (22)$$

Then, based on the denoising function $\epsilon_\theta(S_T, t|X_c)$, the $\mu_\theta(S_t, X_c, t)$ can be calculated as:

$$\mu_\theta(S_t, X_c, t) = \frac{1}{\sqrt{\alpha_t}} \left(S_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(S_T|X_c, t) \right), \quad (23)$$

where $\alpha_t = 1 - \beta_t$, $\bar{\alpha}_t = \prod_{i=0}^T \alpha_i$.

Finally, the reverse denoising process is obtained as:

$$\hat{S}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(S_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(S_T|X_c, t) \right) + \sqrt{\beta_t} z, \quad (24)$$

where $z \sim N(0, 1)$. During the sample generation phase, various samples are drawn from the z distribution at different time intervals, iteratively continuing this process until reaching a predetermined number of steps. The final output, termed $\hat{S}_0$, obtained from the last sampling step.

2) Denoising Network: As illustrated in the bottom-right corner of Fig. 2, our denoising network leverages adaptive modules to integrate multi-scale features extracted by the U-Net [36] and spatial features obtained from the GCN [37], which operates on a dynamically learned graph. The fusion

process begins with feature concatenation along channel dimension, presented as:

$$S_c = \text{Concat}(S_m, S_s), \quad (25)$$

where S_m and S_s are multi-scale features and spatial features, respectively.

To calculate the attention weights, a global average pooling is introduced at the “squeeze” step to generate a channel-wise descriptor, S_f , defined as:

$$S_f = \text{AvgPool2d}(S_c) \quad (26)$$

Next, in the “excitation” step, the squeezed feature descriptor S_f is passed through a 2D convolutional layer (Conv2d) configured to output 2 channels, each corresponding to one of the input feature streams. This produces two distinct raw scores, which are then activated using a Sigmoid activation function (σ) to generate the final attention weights, α_m and α_s . This process can be formulated as:

$$\begin{aligned} S_{\text{score}} &= \text{Conv2d}(S_f), \\ \alpha^{(m)} &= \sigma(S_{\text{score}}^{(m)}), \\ \alpha^{(s)} &= \sigma(S_{\text{score}}^{(s)}), \end{aligned} \quad (27)$$

where $S_{\text{score}}^{(m)}$ and $S_{\text{score}}^{(s)}$ denote the first and second channel, corresponding to the multi-scale and spatial streams, respectively. Finally, the fused output is obtained through element-wise weighted summation of the original feature maps:

$$S_t = \alpha^{(m)} \odot S_m + \alpha^{(s)} \odot S_s, \quad (28)$$

where $\odot$ denotes element-wise multiplication. This adaptive attention mechanism enables the network to dynamically emphasize relevant features from both branches depending on the context.

3) *Anomaly Inference*: Firstly, the error between the reconstructed value $\hat{S}_0$ and the true value S_0 is calculated at time t and sensor i as follows:

$$\text{error}_i^t = |\hat{S}_0 - S_0|. \quad (29)$$

Then, the error values of each sensor are normalized as:

$$E_i(t) = \frac{\text{error}_i - \hat{\mu}_i}{\hat{\sigma}_i}. \quad (30)$$

Finally, we aggregate over sensors using the statistical function to compute the overall anomalousness at time t :

$$DS(t) = \frac{\mu}{\max - \min}, \quad (31)$$

where $\mu = \sum_{i=1}^N E_i(t)/N$, $\max = \max_i E_i(t)$ and $\min = \min_i E_i(t)$. $DS(t)$ is used as the input of SPOT algorithm to automatically identify anomalies [38].

V. EXPERIMENT

A. Dataset

To demonstrate the accuracy and effectiveness of the proposed ICAD method, four publicly available MTS anomaly detection datasets (Swat, WADI, PSM, HAI) and a chemical production process dataset (EO) collected by us are used, which all come from industrial production processes as shown in Table I. The detailed description of these datasets is provided below.

Swat: the secure water treatment dataset, derived from the water treatment simulation test bench, is designed to investigate the launch of network attacks. It records network traffic and readings from 51 sensors and actuators [39].

WADI: serving as an extension of the Swat testbed, the dataset is collected from a water distribution testbed and consists of 16 consecutive days of operations, including 14 routine operations and 2 with attack scenarios [39].

PSM: Pooled Server Metrics dataset collects internally from multiple application server nodes at eBay, which consists of 25 features describing server machine metrics such as CPU utilization and memory. The dataset includes a 13-week training period followed by an eight-week testing phase [40].

EO: The Ethylene Oxide dataset is collected from a real-world, continuous chemical production process characterized by its high-risk and safety-critical nature. The data were recorded in August 2024 at a high sampling frequency of one sample every 5 seconds. It includes measurements from multiple key process units, such as reactors, condensation towers, and distillation towers. To ensure meaningful data representation, we collaborated with experienced frontline operators to select 430 critical variables from thousands of sensors across the plant. These variables include direct physical measurements (e.g., temperature, flow rate, pressure), interlock control variables that trigger automated safety responses, and key indicators used in operational early warning systems. Compared to public benchmarks, EO offers higher dimensionality and richer process semantics, making it a more challenging and realistic testbed.

Anomalies within the dataset are labeled by domain experts and correspond to genuine operational challenges, such as process fluctuations during startup and shutdown phases and abnormal deviations during steady-state operations, providing a reliable ground truth for model evaluation under complex industrial conditions. Due to confidentiality agreements and the proprietary nature of the process, detailed engineering schematics cannot be disclosed. However, we believe that the inclusion of EO as a private, high-fidelity, safety-critical dataset significantly enhances the practical relevance and robustness evaluation of the proposed method.

HAI: (HIL-based Augmented ICS) Security Dataset is collected from a realistic industrial control system (ICS) testbed augmented with a Hardware-In-the-Loop (HIL) simulator that emulates steam-turbine power generation and pumped-storage hydropower generation. Featuring 59 dimensions including temperature, pressure, and flow, this dataset simulates 38 attack types [41].

TABLE II
QUANTITATIVE RESULTS IN TERMS OF PREC, REC, AND F1 SCORE ON FIVE DATASETS WITH GROUND TRUTH LABELLED ANOMALIES.

Method	Swat			WADI			PSM			EO			HAI		
	Rec	Prec	F1	Rec	Prec	F1	Rec	Prec	F1	Rec	Prec	F1	Rec	Prec	F1
One-SVM	0.832	0.7839	0.8072	0.6649	0.7121	0.6877	0.9071	0.7592	0.8266	0.8312	0.5977	0.6954	0.4347	0.7519	0.5509
DSVDD	0.8965	0.7403	0.8109	0.7379	0.7036	0.7203	0.8125	0.9364	0.8701	0.8635	0.6246	0.7249	0.4004	0.7296	0.517
COUTA	0.9365	0.8217	0.8754	0.7217	0.7458	0.7336	0.9417	0.9544	0.948	0.7585	0.7793	0.7688	0.5204	0.7936	0.6286
COCA	0.8946	0.7903	0.8437	0.739	0.7602	0.7495	0.9154	0.9306	0.9229	0.7615	0.7808	0.771	0.6583	0.6964	0.6768
MAD	0.8220	0.7954	0.8085	0.5241	0.6942	0.5973	0.9504	0.8736	0.9104	0.5565	0.8035	0.6576	0.4240	0.5684	0.4857
GDN	0.6812	0.9935	0.8100	0.4019	0.9750	0.5700	0.8372	0.9274	0.8800	0.7510	0.6048	0.6700	0.4545	0.7293	0.5600
GRELEN	0.8350	0.9560	0.8910	0.6130	0.7730	0.6820	0.9420	0.9210	0.9310	0.6490	0.8495	0.7358	0.4852	0.4178	0.4490
GLUE	0.7478	0.9356	0.8312	0.7553	0.5528	0.6384	0.9326	0.8792	0.9051	0.7817	0.7143	0.7465	0.8032	0.6291	0.7056
Lunar	0.8735	0.7369	0.7994	0.5994	<u>0.8905</u>	0.7165	0.8234	0.9071	0.8632	0.6002	0.5891	0.5946	0.4539	0.7126	0.5546
CST_GL	0.8366	0.8706	0.8533	0.2295	0.8389	0.3604	0.8144	0.8703	0.8414	0.4902	0.759	0.5957	0.3955	0.6208	0.4832
LOF	0.9677	0.6563	0.7821	0.8345	0.6218	0.7126	0.9049	0.5789	0.7061	0.7309	0.5056	0.5977	0.4987	0.4271	0.4601
IF	0.7315	0.9620	0.8311	0.6155	0.6241	0.6198	0.9245	0.7609	0.8348	0.6225	0.9451	0.7506	0.5567	0.5340	0.5451
ALAD	0.9314	0.7172	0.8104	0.8397	0.2605	0.3976	0.9719	0.2915	0.4485	0.7750	0.4178	0.5429	0.4973	0.4502	0.4726
OMNIANOMALY	0.8430	0.8142	0.8283	0.9799	0.2652	0.4174	0.7446	0.8839	0.8083	0.8647	0.7063	0.7775	0.4672	0.4835	0.4752
USAD	0.7402	0.9870	0.8460	0.3220	0.6451	0.4296	0.9100	0.6800	0.7800	0.6918	0.6172	0.6524	0.6019	0.6318	0.6165
INTERFUSION	0.8558	0.8059	0.8301	0.7719	0.6842	0.7254	0.8345	0.8361	0.8352	0.6386	0.8041	0.7119	0.4425	0.4672	0.4545
ECOD	0.7771	0.9304	0.8469	0.7340	0.6618	0.6960	0.8633	0.9658	0.9117	0.6225	0.7381	0.6754	0.6921	0.6859	0.6890
A_T	0.9250	0.8049	0.8608	0.3844	0.6885	0.4933	0.9622	0.9313	0.9465	0.8853	0.7085	0.7871	0.6793	0.6658	0.6725
PeFAD	0.9491	0.9119	0.9301	0.9478	0.5107	0.6439	0.9747	0.9799	0.9772	0.7911	0.8273	0.8088	0.9398	0.7256	0.8159
DiffAD	0.9690	0.9844	<u>0.9766</u>	0.8621	0.6078	0.7130	<u>0.9892</u>	0.9700	0.9795	0.8252	0.8086	<u>0.8168</u>	0.8290	0.8372	<u>0.8331</u>
DDMT	0.9978	0.9656	0.9724	0.7153	0.7821	0.7472	0.9840	<u>0.9792</u>	0.9816	0.6793	0.8124	0.7399	0.7763	<u>0.8489</u>	0.8110
ImDiffusion	0.8465	0.8988	0.8709	0.7596	0.7315	0.7453	0.9753	0.9811	0.9781	0.7225	0.7873	0.7535	0.6192	0.6923	0.6537
Diffstg	0.9037	0.9492	0.9259	0.7125	0.8366	<u>0.7696</u>	0.9215	0.9379	0.9296	0.7405	0.8312	0.7832	0.8003	0.8459	0.8225
ICAD (ours)	<u>0.9889</u>	0.9794	0.9841	<u>0.9264</u>	0.7834	0.8489	0.9985	0.9763	0.9873	0.8977	0.8400	0.8679	<u>0.8591</u>	0.8872	0.8730

TABLE I
DATA INFORMATION ABOUT DIMENSIONS, THE SIZE OF TRAINING, VALIDATION AND TESTING, ANOMALY RATIO.

Datasets	No. of dimensions	Training set size	Validation set size	Testing set size	Anomaly ratio(%)
EO	430	1632	408	2040	12.21
Swat	51	396,000	99,000	449,919	11.97
WADI	118	967,680	241,921	1,728,01	5.71
PSM	25	105,984	26,497	87,841	27.76
HAI	59	192,960	48,240	153,000	3.91

B. Baselines and Evaluation Metrics

We evaluate the ICAD algorithm against 23 baselines across five datasets using popular performance metrics: precision (Prec), recall (Rec), F1, AUPRC and AUROC. Specifically, these algorithms are divided into four types, graph-based learning, non-graph learning, recently popular diffusion model and one class learning.

The evaluated graph-based learning methods include:

- MAD (ICDM 2020) constructs two graph attention layers to learn the complex dependencies of MTS from time and feature dimensions [42].
- GDN (AAAI 2021) uses attention mechanism to learn the topology of graph neural networks (GNN) to detect anomalies [5].
- GLUE (2021) uses graph deviation network with local uncertainty estimation to learn complex dependencies between variables [43].
- GRELEN (IJCAI 2022) combines graph neural network (GNN) and stochastic graph relational learning strategy to capture the between-sensor dependence [8].
- Lunar (AAAI 2022) learns to use information from the nearest neighbors of each node to find anomalies [6].
- CST-GL (TNNLS 2024) captures spatial-temporal dependencies via dynamic correlation learning, multi-hop graph convolutions, and dilated temporal modeling [49].

dencies via dynamic correlation learning, multi-hop graph convolutions, and dilated temporal modeling [49].

The selected non-graph learning methods are as follows:

- LOF (SIGMOD) calculates an outlier factor for each data point by defining the locally accessible density, which identifies the outlier degree of a data point [25].
- IF (ICDM 2008) aims to find anomalies at leaves that are relatively close to the root of a decision tree, i.e., at a shallower depth of a decision tree [27].
- ALAD (ICDM 2018) uses adversarially learned features by bi-directional generative adversarial networks to detect anomaly sample [10].
- OMNIANOMALY (KDD 2019) aims to capture the normal patterns of MTS by learning their robust representations with key techniques such as stochastic variable connection and planar normalizing flow [9].
- USAD (KDD 2020) learns a robust representation for the raw time-series input using an adversarially trained encoder-decoder pair and the reconstruction error is used as the anomaly score [4].
- INTERFUSION (KDD 2021) aims to model the normal patterns inside MTS data through hierarchical variational autoencoder with two stochastic latent variables [7].
- ECOD (TKDE 2022) computes an empirical cumulative distribution along each data dimension, and then utilizes this distribution to estimate the tail probability [28].
- ANOMALY TRANSFORMER (ICLR 2022) aims to compute the association discrepancy by a new Anomaly Attention mechanism. For the sake of clarity in writing, this method is designated as “A_T” [30].
- PeFAD (KDD 2024) uses pre-trained language models and parameter-efficient tuning to detect anomalies [48].

TABLE III
QUANTITATIVE RESULTS IN TERMS OF AUPRC AND AUROC SCORE ON FIVE DATASETS WITH GROUND TRUTH LABELLED ANOMALIES.

Method	Swat		WADI		PSM		EO		HAI		Average	
	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC	AUPRC	AUROC
One-SVM	0.8153	0.9015	0.6648	0.7743	0.8211	0.9058	0.5490	0.7985	0.5271	0.7594	0.6755	0.8279
DSVDD	0.8007	0.9174	0.6832	0.8049	0.8351	0.9210	0.5615	0.8134	0.4639	0.7135	0.6689	0.8340
COUTA	0.8949	0.9610	0.7517	0.9295	0.9391	0.9538	0.8360	0.9158	0.7845	0.8820	0.8412	0.9284
COCA	0.8126	0.8779	0.6572	0.8459	0.8893	0.9420	0.7552	0.8736	0.6922	0.8716	0.7613	0.8822
MAD	0.8245	0.9133	0.6635	0.8244	0.8705	0.9218	0.7006	0.7995	0.5131	0.7996	0.7144	0.8517
GDN	0.8403	0.9036	0.4192	0.6633	0.8596	0.9054	0.6493	0.7377	0.7434	0.9047	0.7024	0.8229
GRELEN	0.8692	0.9764	0.6548	0.8032	0.8932	0.9634	0.7296	0.8577	0.4288	0.7632	0.7151	0.8728
GLUE	0.8126	0.8693	0.7503	0.8180	0.8516	0.9050	0.7517	0.8283	0.7644	0.8569	0.7861	0.8555
Lunar	0.7768	0.8460	0.7764	0.8757	0.8845	0.9166	0.7104	0.8056	0.5779	0.8004	0.7452	0.8489
CST_GL	0.8469	0.8935	0.2011	0.6933	0.8536	0.9155	0.5289	0.7834	0.4164	0.8034	0.5694	0.8178
LOF	0.7245	0.8967	0.6483	0.7954	0.7049	0.819	0.5632	0.7854	0.4537	0.6858	0.6189	0.7965
IF	0.7343	0.8768	0.4249	0.8337	0.8169	0.9006	0.648	0.8039	0.4777	0.7404	0.6204	0.8311
ALAD	0.7945	0.9032	0.2197	0.7632	0.3638	0.853	0.4582	0.841	0.4038	0.8057	0.4480	0.8332
OMNIANOMALY	0.7984	0.8946	0.2187	0.7853	0.8426	0.9034	0.7642	0.8953	0.3692	0.8058	0.5986	0.8569
USAD	0.8545	0.9077	0.3098	0.7631	0.7955	0.884	0.5195	0.7746	0.5944	0.864	0.6147	0.8387
INTERFUSION	0.707	0.8489	0.5396	0.8154	0.804	0.8981	0.7022	0.7937	0.4245	0.6985	0.6355	0.8109
ECOD	0.7507	0.8846	0.5009	0.857	0.8717	0.9258	0.5055	0.7958	0.664	0.8254	0.6586	0.8577
A_T	0.8973	0.9564	0.4579	0.7654	0.9201	0.9385	0.6893	0.8007	0.7643	0.8955	0.7458	0.8713
PeFAD	0.9274	0.9682	0.5424	0.7916	0.9631	0.9742	0.7393	0.8687	0.6842	0.8131	0.7713	0.8832
DiffAD	0.9752	0.9952	0.7144	0.9033	0.9429	0.9677	0.8588	0.9300	0.8795	0.9233	0.8742	0.9439
DDMT	0.9803	0.9916	0.7526	0.8904	0.9594	0.9906	0.8629	0.9246	0.8406	0.8947	0.8792	0.9384
ImDiffusion	0.8226	0.8763	0.7380	0.8776	0.9699	0.9780	0.7842	0.8782	0.5139	0.7785	0.7657	0.8777
Diffstg	0.9164	0.9580	0.7699	0.8837	0.9338	0.9652	0.8470	0.9188	0.8571	0.9120	0.8648	0.9275
ICAD (ours)	0.9817	0.9935	0.8328	0.9667	0.9652	0.9811	0.9046	0.9753	0.9154	0.9855	0.9199	0.9804

For diffusion-based learning, we compare our method with:

- DiffAD (2023 KDD) enhances the imputation performance of missing values with conditional weight-incremental diffusion by a new denoising diffusion-based imputation method [13].
- DDMT (2023) integrates a novel adaptive dynamic neighbor mask mechanism with the Transformer and Denoising Diffusion Model for anomaly detection [14].
- ImDiffusion (VLDB 2024) combines time series imputation and diffusion models to achieve accurate and robust anomaly detection [15].
- Diffstg (SIGSPATIAL 2023) combines the spatio-temporal learning capabilities of spatio-temporal graph neural networks with the uncertainty measurements of diffusion models [32].

The one-class learning baseline is represented by:

- One-SVM (NIPS) identifies anomalies by finding a hyperplane that best separates the majority of the data points from the origin in a high-dimensional feature space [44].
- DSVDD (ICML 2018) uses deep learning to map data into a new space where normal data points cluster tightly, allowing effective identification of anomalies [45].
- COUTA (TKDE 2024) realizes contamination-tolerant, anomaly-informed learning of data normality via uncertainty modeling-based calibration and native anomaly-based calibration [31].
- COCA (ICDM 2023) follows the normality assumptions of contrastive learning (CL) and one-class classification. It treats the original and reconstructed representations as the positive pair of negative-sample-free CL [46].

C. Overall Performance

According to the results in Table II, Table III and Fig. 3, a comprehensive analysis of the performance of our proposed

ICAD method leads to the following conclusions:

- ICAD consistently outperforms the state-of-the-art baseline methods on all datasets. It demonstrates that ICAD is able to learn the logical relationship between the trend-pattern cues and the contextual window of time series to improve the forecasting performance.
- In comparisons between six graph-based learning methods and nine non-graph-based approaches, it is clear that graph learning techniques consistently outperform non-graph counterparts, particularly on datasets with complex topologies such as WADI, EO, and HAI. This performance gap is especially evident in tasks that require the capture of intricate relationships among data points, which are crucial for accurate predictions. ICAD's ability to effectively model these relationships enables it to achieve superior performance in such tasks, highlighting its capability to handle complex and diverse structures.
- Comparing diffusion models with other approaches, it is found that diffusion models offer significant advantages due to their diverse generative capabilities and rigorous mathematical foundations. However, current diffusion models still lack targeted cue guidance, which can lead to challenges in achieving high fidelity in practical applications. The proposed ICAD method can significantly enhance the robustness of diffusion model by incorporating trend textual description cues.
- Despite the increased complexity in the HAI, WADI, and EO datasets, characterized by more sensor variables and intricate anomaly patterns, Fig. 3 demonstrates that ICAD consistently outperforms SOTA methods, maintaining remarkable stability even in large-scale deployments with up to 430 sensors. Specially, the uptime of the algorithm and its advantages in actual IIoT deployment are described in Subsection 5.7.

TABLE IV
PERFORMANCE IMPROVEMENT OF F1 IN FIVE DATASETS.

Method	Datasets					
	PSM	Swat	HAI	WADI	EO	AVG
SOTA	0.9816	0.9766	0.8331	0.7696	0.8168	0.8755
ICAD	0.9873	0.9841	0.8730	0.8489	0.8679	0.9122
Improvement	0.58%↑	0.77%↑	4.79%↑	10.30%↑	6.26%↑	4.54%↑

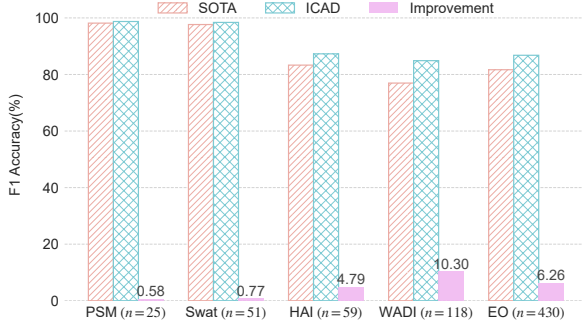


Fig. 3. The performance of algorithms on different complex datasets. The SOTA represents the optimal baseline algorithm.

- ICAD achieves the highest average AUPRC (0.9199) and the highest average AUROC (0.9804), confirming its robust and superior performance. ICAD consistently attains the highest AUPRC score on all five datasets, which is particularly significant as AUPRC is widely regarded as a more informative metric than AUROC for heavily imbalanced tasks. This superiority is especially evident on the challenging HAI and WADI datasets, which feature highly imbalanced distributions. On these difficult benchmarks where many competing methods struggle, ICAD achieves exceptional AUPRC scores of 0.9154 and 0.8328 respectively, establishing a clear advantage over other methods.

To provide a more intuitive comparison between our proposed ICAD algorithm and the current state-of-the-art algorithms for each dataset, we present the performance improvements in F1 scores across five datasets (see Table IV). The following observations can be made:

- On the PSM and Swat datasets, ICAD shows a slight performance improvement of 0.58% and 0.77%, respectively, due to the relative simplicity of these datasets. In contrast, on the WADI and EO datasets, the ICAD algorithm exhibits significant improvements. This enhancement is attributed to the ICAD algorithm's ability to capture complex trend patterns and uncertainties in the data, improving the model's adaptability to complex scenarios.
- On average, our ICAD algorithm outperforms SOTA methods overall, achieving a 4.54% increase. This improvement is significant for the intelligent monitoring of actual production processes, as it enhances the algorithm's ability to detect anomalies, predict potential failures, and optimize operational efficiency.

TABLE V
ABLATION STUDIES

Method	Datasets				
	Swat	WADI	PSM	EO	HAI
w/o cue	0.9748	0.7866	0.9542	0.8121	0.8265
w/o relation	0.9576	0.8135	0.9469	0.8236	0.8428
w/o fusion	0.9602	0.8092	0.9711	0.8306	0.8319
W/ complex	0.9886	0.8605	\	\	0.8827
W/ word2Vec	0.9788	0.8275	0.9792	0.8421	0.8652
W/ cluster	0.9863	0.8302	0.9890	0.8424	0.8612
W/ mlp	0.9703	0.8145	0.9762	0.8412	0.8497
ICAD	0.9841	0.8489	0.9873	0.8679	0.8730

D. Ablation Analysis

We conduct ablation studies to validate the effectiveness of our key components that contribute to the improvements on five datasets. We compare ICAD with the following variants:

- *w/o cue*: This variant does not use textual embeddings to describe trend cues as conditions for a diffusion model, i.e. an unconditional diffusion model.
- *w/o relation*: This variant does not take into account the logical relationship between time series windows, implying a purely conditional diffusion model.
- *w/o fusion*: This variant uses the U-net network directly as a denoising network, ignoring multiple features and adaptive fusion modules.
- *W/ complex*: For datasets with available metadata (Swat, WADI, HAI), we replaced generalized trend descriptions with detailed prompts incorporating equipment and sensor semantics. These prompts incorporate specific metadata such as equipment names and sensor tags (e.g., Temperature sensors in unit P-101 exhibit a sharp increase). For datasets like PSM and our proprietary EO, which do not disclose specific sensor semantics, constructing these detailed prompts is infeasible, marked as “\”.
- *W/ word2Vec*: We replaced the BERT encoder with a standard Word2Vec model to generate text embeddings, while keeping the rest of model architecture unchanged.
- *W/ cluster*: Training-time windows are clustered, and each resulting cluster is labeled based on its dominant trend pattern (e.g., U-shaped, monotonic increase). During inference, each new time series window is assigned to the nearest cluster, and the corresponding label is used as the trend cue.
- *W/ mlp*: We replaced the Kalman filter module with a standard multi-Layer perceptron (MLP).

Table V shows the F1 accuracy of different variants. The results reveal the following key insights:

(1) The textual embeddings for trend cue descriptions are vital, especially for datasets with complicated trends, such as WADI, EO, and HAI, which involve industrial operations like shutdowns and startups. This finding aligns with our expectations, as these embeddings guide the diffusion model to generate high-fidelity time series, crucial for capturing complex patterns in challenging datasets. (2) Existing conditional diffusion models often overlook the contextual relationships between windows, leading to overfitting and suboptimal performance. (3) Simply using a U-Net network is insufficient

for capturing the spatial features of time series data. In contrast, an adaptive fusion module with GCN designed by us enhances the model's ability to extract and generalize across multiple features. (4) The W/ complex variant achieves state-of-the-art performance on the three datasets where it could be applied. This result strongly suggests that incorporating rich, domain-specific knowledge can significantly enhance model precision. However, the primary limitation of this approach is its dependency on detailed metadata, which is often proprietary or unavailable in many real-world datasets, as demonstrated by our inability to apply it to PSM and EO. In contrast, our proposed ICAD, using generalized trend cues, delivers consistently high and robust performance across all five datasets, including those where detailed descriptions are not feasible. This confirms ICAD's robustness and practical value. (5) Our proposed ICAD method, which utilizes a BERT, consistently outperforms the Word2Vec-based variant across all five datasets. This result strongly suggests that the contextualized and representations generated by BERT are critical for effectively conditioning the diffusion process. The ability of BERT to understand the semantics of the trend descriptions leads to more precise guidance and, ultimately, superior performance. (6) This W/ cluster variant performs reasonably on some datasets (e.g., PSM), but it is notably outperformed by ICAD on more complex industrial datasets such as WADI, EO, and HAI. While clustering provides a reasonable approximation, its performance is limited by the difficulty of forming distinct and semantically meaningful clusters in high-dimensional, noisy data. In contrast, our direct, mathematically-principled method, identifying trend patterns individually for each window, proves to be more robust and avoids potential error propagation from an intermediate, imperfect clustering mathematically-principled calculation is applied to each window individually. (7) Across all datasets, our Kalman filter-based ICAD consistently and significantly outperforms the MLP-based alternative. This provides strong empirical evidence that the structured, probabilistic modeling of dynamics in Kalman filtering provides significant advantages over purely learned embedding, particularly in noisy or complex environments.

E. Effect of Parameters

We evaluate the impact of the window w on the performance of ICAD. The results, shown in Fig. 4, indicate that ICAD can achieve stable results under different parameter settings. The stability is attributed to model's ability to leverage the relationships between contextual windows to train the diffusion model, effectively mitigating the impact of window size on model performance. By incorporating contextual dependencies, the model can better capture underlying trend patterns within time series data, leading to improved accuracy.

F. Visualization of Difference histogram and Uncertainty

To visualize the uncertainty of the trend, we convert the time series into a histogram. This transformation allows for a better understand and analysis of the distribution and variation of data from a new perspective, enabling effective identification of

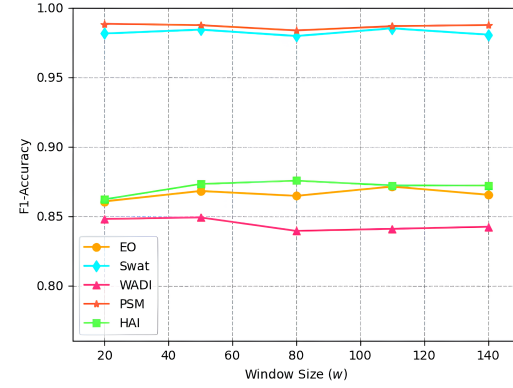


Fig. 4. F1 score of ICAD with different window size w .

Algorithm 1 Training the Condition Diffusion Model

Input: noise level $N \in \mathbb{N}^+$, data $X^0 \sim q(X^0)$, $X^0 \in \mathbb{R}^{M \times N}$, Kalman filter K .
repeat
State vector $S^0 = K(X^0)$
Difference vector $x_d = \text{diff}(X^0)$
The trend pattern f obtained by Eq.(6)-Eq. (10)
and uncertainty obtained by Eq. (11)-Eq. (13)
Text: the trend pattern and uncertainty are $\{\}$ and $\{\}$.
Initialize $t \sim \text{Uniform}(1, \dots, N)$ and $\epsilon \sim \mathcal{N}(0, I)$
The condition $X_C = \text{MLP}(\text{Bert}(\text{Text}))$
Compute the loss function for the input X^n
(S^0 after t steps of noise):

$$L_{dif} = \|\epsilon - \epsilon_\theta(\sqrt{\alpha_t}S^0 + \sqrt{1 - \alpha_t}\epsilon | X_C, t)\|^2$$

The inference loss function L_{reg} obtained by Eq. (19).
Take gradient step on $\nabla_\theta(L_{dif} + L_{reg})$
until converged
Output: $\epsilon_\theta(\bullet)$

abnormal or unusual behavior. In particular, by comparing the histograms in different time windows, we can indirectly observe the trend changes and periodic patterns of the time series. For example, periodic changes may manifest as fluctuations in data frequency within specific intervals on the histogram.

According to Fig. 5, we observe the following:

- The time series data in window 1 (subfigure (e)) and window 2 (subfigure (f)) both generally exhibit a U-shaped trend. However, window 2 shows a notable decline in the middle, indicating increased uncertainty and resulting in a lower overall information entropy. Additionally, the histogram for window 2 (subfigure (b)) shows a clear concentration of values in the middle range, highlighting the effectiveness of the difference histogram in capturing this sudden decline in the trend.
- The time series in window 3 (subfigure (c)) and window 4 (subfigure (d)) follow the same monotonic trend, with comparable levels of uncertainty and overall information entropy. This consistency demonstrates the effectiveness of our ICAD algorithm in measuring trend uncertainty. Additionally, the difference histogram clearly shows that,

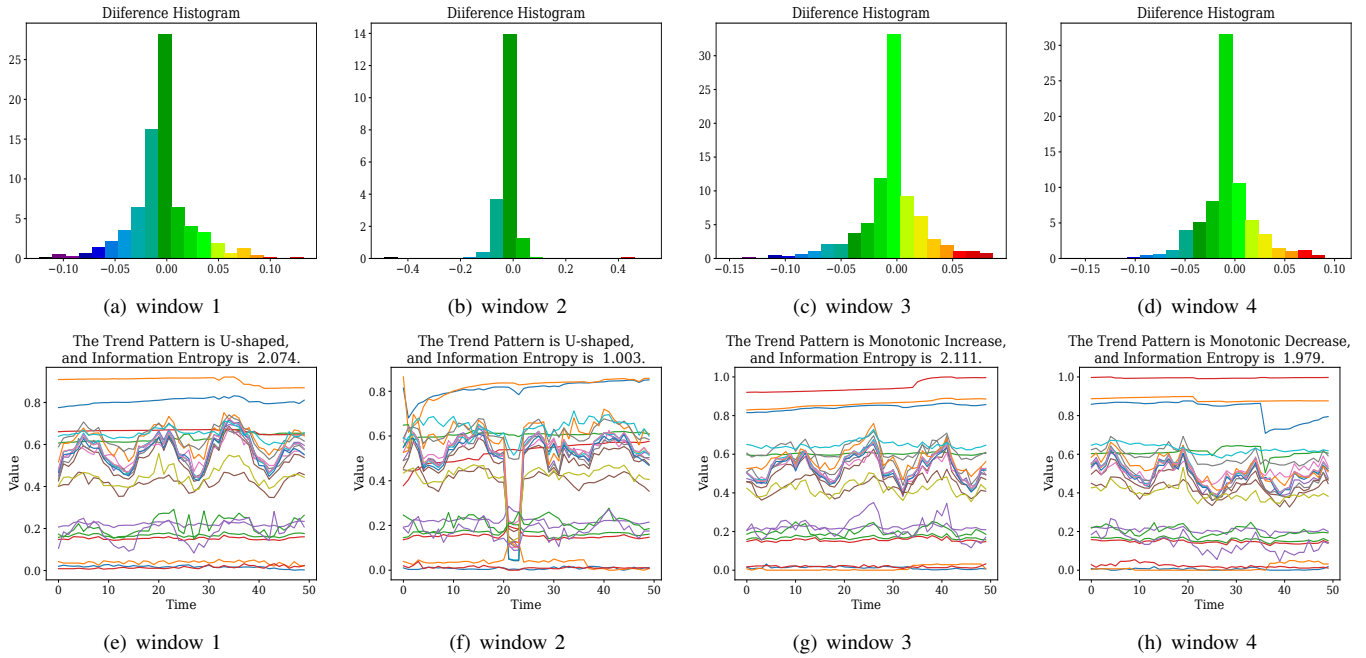


Fig. 5. Visual demonstration of the trend cue extraction process detailed in Section IV.A. The bottom row (e)-(h) displays specific time series windows (showing all 25 sensor variables from the PSM dataset), along with the automatically calculated Trend Pattern and Information Entropy. The top row (a)-(d) displays the corresponding histograms of the flattened first-order differences, which are the basis for the entropy calculation. The colors in the histograms are utilized to visually emphasize the structural differences and spread of the data distributions across these windows.

Algorithm 2 Anomaly Inference

Input: noise level $M \in \mathbb{N}^+$, test data $X^0 \in \mathbb{R}^{M \times N}$, Kalman filter K .

repeat

State vector $S^0 = K(X^0)$

Difference vector $x_d = \text{diff}(X^0)$

The trend pattern f obtained by Eq.(6)-Eq. (10)

and uncertainty obtained by Eq. (11)-Eq. (13)

Text: the trend pattern and uncertainty are $\{\}$ and $\{\}$.

$X_C = \text{MLP}(\text{Bert}(\text{Text}))$

for $t = M, \dots, 1$ do

$z \sim \mathcal{N}(0, I)$ if $t > 1$ else $z = 0$

$\tilde{S}^{t-1} = \frac{1}{\sqrt{\alpha_t}}(\tilde{S}^t - \frac{1-\alpha_t}{\sqrt{1-\alpha_t}}\epsilon_\theta(\tilde{S}^t|X_C, t)) + \sqrt{1-\alpha_t}z$

$E = \|\tilde{S}^0 - S^0\|^2$

end

$Y = 1$ if $E \geq \tau$ else -1, where τ is anomaly score threshold, computed using the SPOT [38].

end

Output: anomaly labels Y

despite the similar shapes, the locations of the highest value sets differ between the two windows, indicating ICAD's ability to effectively distinguish between upward and downward trends.

G. Running Settings, Time, and Memory

To evaluate the computational cost of both training and inference on five datasets, as shown in Table V, we conduct experiments with the sampling steps of the diffusion model



Fig. 6. GPU Memory Utilization for model training.

set to 100 for all datasets. These tests are crucial due to the high time requirements for deployment in actual IIoT, where efficient training and inference are essential to ensure timely and reliable application of our proposed ICAD algorithm. To further address edge deployment feasibility, we explicitly monitor the GPU memory footprint during execution, and visualize the peak usage across datasets in Fig. 6. The experiments are executed on a single i7-8700 CPU @ 3.20GHz and an NVIDIA GeForce RTX 3080 Ti GPU. The window size (w), number of epochs $epochs$, and number of hidden layers h for the five datasets (EO, Swat, WADI, PSM, and HAI) are 50/200/32, 50/200/32, 50/50/32, 50/300/16, and 50/400/32, respectively.

From Table VI and Fig. 6, the following conclusions can be drawn:

- The training time of our ICAD algorithm is influenced

TABLE VI
THE COMPUTATIONAL COST IN BOTH TRAINING AND INFERENCE ON SWAT, WADI, PSM, EO AND HAI DATASETS. THE UNIT OF RUNNING TIME IS SECONDS.

	Swat		WADI		PSM		EO		HAI	
	training	inference	training	inference	training	inference	training	inference	training	inference
all time	5373.39	674.4	6497.38	745.69	5135.55	196.70	130.01	42.78	6284.35	559.42
epoch size	200	\	50	\	300	\	200	\	400	\
batch size	16	16	16	16	16	16	16	16	16	16
window size	50	50	50	50	50	50	50	50	50	50
window_number	9900	6898	24192	3456	2595	1756	48	48	6192	5832
each window time	0.54	0.10	0.27	0.22	2.00	0.11	2.70	0.89	1.01	0.10

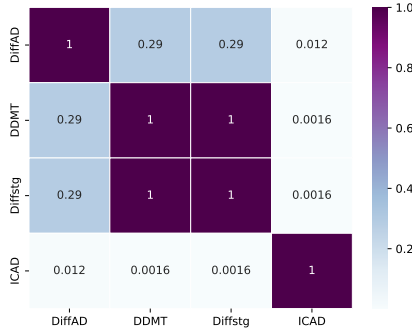


Fig. 7. P value matrix of F1 for graph-based method in five datasets, each value in the figure represents the degree of performance difference between the corresponding algorithms. Smaller values indicate more obvious differences between the algorithms, while larger values signify less obvious distinctions.

by the number of epochs, the number of sensors in each dataset, and the complexity of data anomalies, resulting in an overall efficient training duration. Despite using a single NVIDIA GeForce RTX 3080 Ti, the longest training time is for the WADI dataset, taking only 0.3 hours per 50 epochs. If the graphics card with the computing power of A100 or A800 is used, the training time of our algorithm would be significantly reduced, making it more suitable for the actual IIoT.

- Our proposed ICAD algorithm consistently achieves an inference time of less than 1.01 seconds per window on a single GPU across all five datasets, highlighting its efficient memory usage and rapid computation capabilities. This efficiency indicates that the ICAD algorithm is well-suited to meet real-time detection requirements. The quick inference time allows the ICAD algorithm to analyze incoming data streams and identify anomalies swiftly, minimizing delays. Such performance is critical in applications where the timely detection of anomalies is paramount, including industrial monitoring, cybersecurity, and healthcare. In these domains, the ability to promptly respond to anomalies can prevent system failures, enhance security measures, and improve patient outcomes by enabling proactive interventions.
- The peak allocated memory during training is 4.16 GB, with the average memory usage being only 1.00 GB. While a 4.16 GB peak training memory might be too high for highly constrained microcontrollers at the extreme

edge, it is perfectly feasible for deployment on more capable industrial edge servers, gateways, or local control room machines.

A central consideration during our design process was to ensure the models computational efficiency for practical deployment. To mitigate the resource-intensiveness typically associated with multi-module deep learning models, we implemented several strategic architectural choices:

Compact BERT: We utilize a foundational, compact version of BERT (bert-base-uncased) rather than larger, more complex BERT variants, to strike a balance between expressive power and computational cost.

Minimalist GCN Architecture: GCN is intentionally designed with only one layer, sufficient to capture essential spatial correlations between sensors while significantly reducing the computational cost associated with deeper graph networks.

U-Net Simplification: The attention mechanism is removed from U-Net architecture, balancing multi-scale feature extraction with computational efficiency.

H. Generalization and Robustness

1) *Generalization Performance Test:* To comprehensively compare the generalization performance of different learning algorithms, it is insufficient to rely solely on evaluating their performance on individual datasets. Hypothesis testing provides a crucial foundation for such comparisons. In this paper, we employ the Friedman test and the Conover post hoc test [47] to evaluate the performance of multiple algorithms across different datasets. Since the F1 score serves as an average measure of both recall (Rec) and precision (Prec), our analysis focuses exclusively on the F1 score. The generalization performance evaluation is divided into two stages:

In the first stage, Friedman test is conducted under the null hypothesis that there is no difference in performance among algorithms. The calculated F-statistic (Eq. (32)) is 14.106843, which exceeds the critical value of 3.076797 at a confidence level of 0.0001. Therefore, the null hypothesis is rejected, indicating significant differences among the 23 algorithms.

$$T_F = \frac{(N-1)\tau_{\chi^2}}{N(k-1) - \tau_{\chi^2}}, \quad (32)$$

$$\tau_{\chi^2} = \frac{12N}{k(k-1)} \left(\sum_{i=1}^k r_i^2 - \frac{k(k+1)}{4} \right), \quad (33)$$

where k and N represent the number of methods and datasets, and r_i is the average sorting value of the i th algorithm.

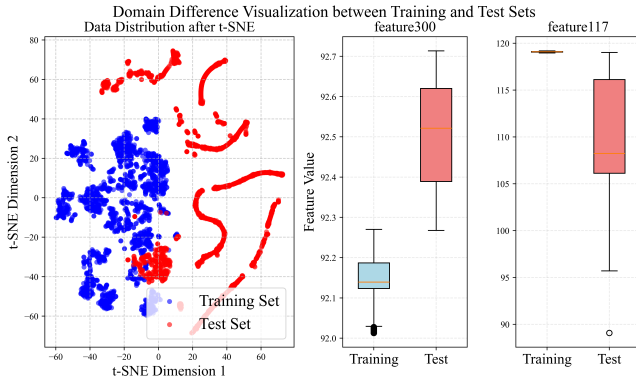


Fig. 8. Domain discrepancy visualization for the EO dataset.

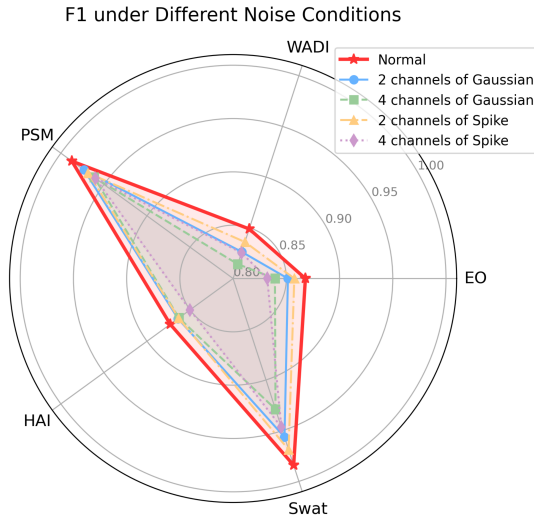


Fig. 9. Robustness evaluation of the model against different types and levels of noise across five benchmark datasets.

In the second stage, after confirming the differences among the algorithms, the Conover post hoc test is conducted to compare whether there are significant differences between pairs of algorithms. At the 95% confidence level, the proposed ICAD algorithm is significantly different from all other algorithms except for DiffAD, DDMT, and Diffstg.

To further explore the differences among our proposed ICAD algorithm and the DiffAD, DDMT, and Diffstg algorithms, a Conover post hoc test is conducted specifically for these four algorithms. As shown in Fig. 7, at the 95% confidence level, our proposed ICAD algorithm is more effective than the other three algorithms.

2) *Robustness Analysis*: A key strength of our proposed method is its robustness to both high-level domain shifts and data perturbations. We validate this through two targeted sets of experiments.

First, to evaluate robustness under domain shift, we conduct a case study using the EO dataset, which exhibits a clear distributional shift between the training and test sets. As shown in Fig. 8, a t-SNE visualization reveals a distinct separation between the training (blue) and test samples (red). Despite this, our model achieves a state-of-the-art F1-score on the EO

test set. This robustness stems from the fact that our trend and uncertainty cues are not based on fixed or global templates. Instead, they are dynamically calculated for each time series window based on difference statistics and information entropy. This allows the conditioning of our diffusion model to adaptively respond to shifts in data distribution, making it resilient to overarching domain shifts.

Second, we conduct additional experiments to explicitly evaluate the sensitivity of our method to sensor noise and outliers. Two types of perturbations were introduced into the test sets: Gaussian noise added to 2 or 4 randomly selected sensor channels, and sporadic, sharp spike outliers added to 2 or 4 randomly selected channels. As Fig. 9 demonstrates, our model's performance remains highly stable. With only minor degradation as noise levels increase, the F1-scores remain very strong across all datasets. This resilience stems from the fact that our trend and uncertainty calculations are based on statistics aggregated over a time window. This aggregation effectively smooths out transient anomalies and reduces the influence of localized sensor noise, leading to a stable and reliable conditioning signal for the diffusion model.

VI. CONCLUSION

In this work, we propose a novel unsupervised anomaly detection method, termed ICAD, which uses defined textual descriptions of trend patterns and uncertainties as conditions. To enable the diffusion model to more accurately capture the logical relationships between contextual windows, a reasoning network is introduced. In particular, a multi-feature adaptive fusion network is designed to enhance the ability of diffusion model to predict noise. Experiments conducted on five real-world industrial datasets demonstrate that ICAD method outperforms baseline methods in both accuracy and generalization. Such a study provides insights for industrial operators and engineers, enabling more effective monitoring of production states and the rapid identification of abnormal behavior. In turn, this facilitates timely corrective actions and drives improvements in IIoT.

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